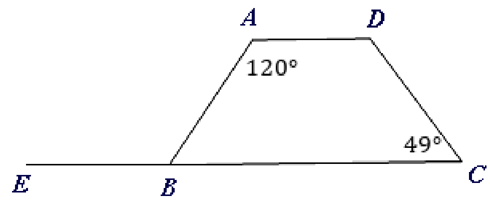
Properties of trapezoids

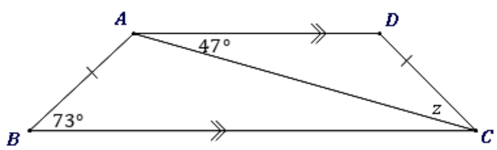
1 Consider the diagram shown, given that $ABCD$ is a trapezoid.

- **a** Identify which sides are parallel.
- **b** Find $m\angle ABC$.
- **c** Find $m\angle ADC$.

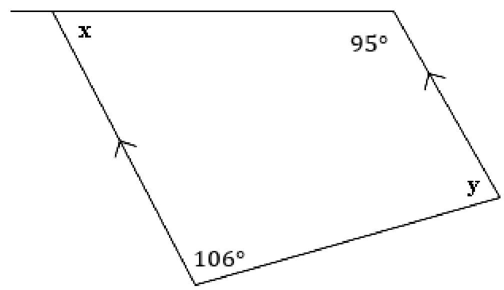


2 Find the value of the variable(s) in the following diagrams.

a



b



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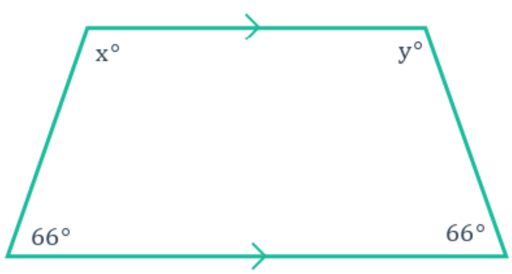
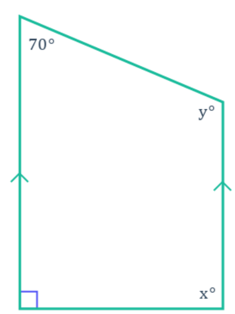
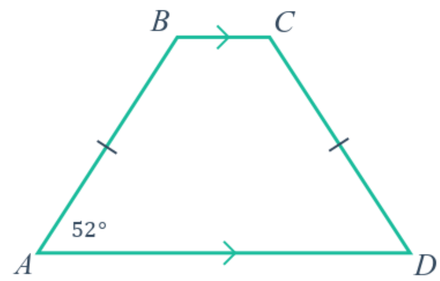


d



3 Consider the following trapezoid.

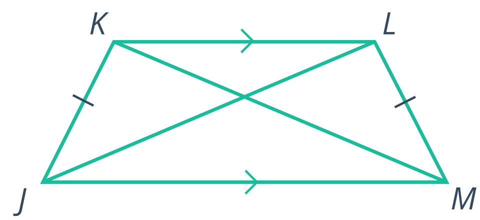


a Find the value of $m\angle B$.

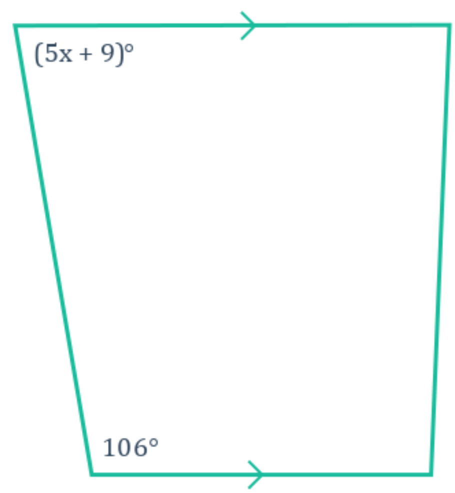
b Find the value of $m\angle C$.

c Find the value of $m\angle D$.

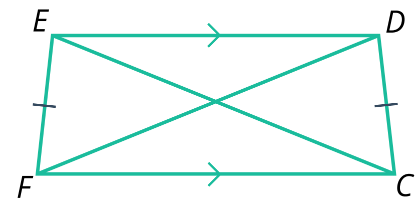
4 Consider the following trapezoid $JKLM$, where $KM = 20$.
Find the value of JL.



5 Consider the following trapezoid.
Find the value of x.



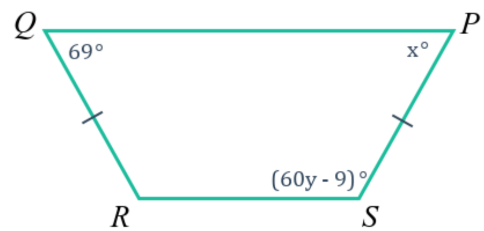
6 Consider the following trapezoid $CDEF$.
For this trapezoid, $EC = 21$ and
$FD = 6x - 9$.
Find the value of$x$.



7 Consider the following trapezoid.

a Find the value of x.

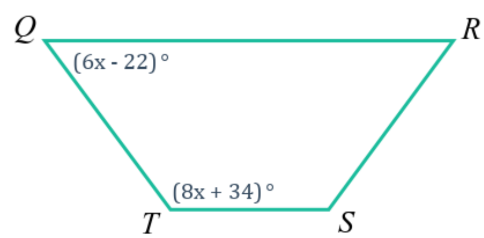
b Find the value of y.



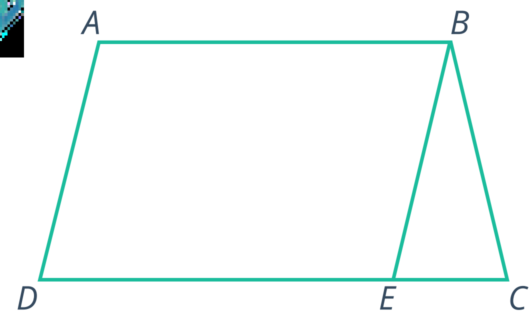
8 Consider the following trapezoid.

a Find the value of x.

b Find $m\angle Q$.



9 Given: $\overline{AB} \parallel \overline{DC}$, $\overline{AD} \cong \overline{BC}$, $\overline{AD} \parallel \overline{BE}$
Prove: $\angle BCE \cong \angle ADE$ and
$\angle BAD \cong \angle ABC$
Note that D, E, and C are collinear.



To prove: $\angle BCE \cong \angle ADE$ **and** $\angle BAD \cong \angle ABC$

	Statements	Reasons
1.	$\overline{AB} \parallel \overline{DC}$, $\overline{AD} \cong \overline{BC}$, $\overline{AD} \parallel \overline{BE}$	Given
2.	$ABED$ is a parallelogram	Definition of a parallelogram
3.	$\overline{AD} \cong \overline{BE}$	Opposites sides of a quadrilateral are congruent
4.	$\overline{BE} \cong \overline{BC}$	Transitive property of congruence
5.	$\triangle EBC$ is isosceles	Definition of an isosceles triangle
6.	$\angle BCE \cong \angle BEC$	Base angles of an isosceles triangle are congruent
7.	$\angle BEC \cong \angle ADE$	Corresponding angles theorem
8.	$\angle BCE \cong \angle ADE$	Transitive property of congruence
9.	$\angle BCE$ and $\angle ABC$ are supplementary	Consecutive interior angles postulate
10.	$\angle ADE$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
11.	$\angle BAD \cong \angle ABC$	Congruent supplements theorem

Given the proof above, determine whether the following statements are true.

a If a trapezoid is isosceles, then base angles are congruent.

b If a trapezoid has one pair of congruent base angles, then it is isosceles.

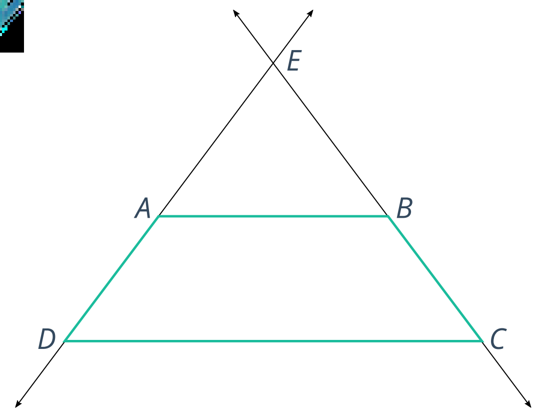
c If a quadrilateral is a trapezoid, then base angles are supplementary.

d If a quadrilateral has one pair of supplementary base angles, then it is a trapezoid.

10 Given:

- $ABCD$ is a trapezoid with bases $\overline{AB}$ and $\overline{DC}$
- $\angle BCD \cong \angle ADC$
- $\overleftrightarrow{AD}$ and $\overleftrightarrow{BC}$ intersect at E

Prove: $ABCD$ is an isosceles trapezoid



To prove: $ABCD$ is an isosceles trapezoid

	Statements	Reasons
1.	$ABCD$ is a trapezoid	Given
2.	$\angle BCD \cong \angle ADC$	Given
3.	$\overline{AB} \cong \overline{DC}$	Definition of a trapezoid
4.	$\angle EAB \cong \angle EDC$ and $\angle EBA \cong \angle ECD$	Corresponding angles theorem
5.	$\angle EAB \cong \angle EBA$	Transitive property of congruence
6.	$\triangle EAB$ is isosceles and $\triangle EDC$ is isosceles	If a triangle has congruent base angles, then it is isosceles.
7.	$\overline{AE} \cong \overline{BE}$ and $\overline{DE} \cong \overline{CE}$	Definition of an isosceles triangle
8.	$AE = BE$ and $DE = CE$	Definition of congruence
9.	$DA + AE = DE$ and $CB + BE = CE$	Segment addition postulate
10.	$DA = DE - AE$ and $CB = CE - BE$	Subtraction property of equality
11.	$DA = CB$	Substitution
12.	$\overline{DA} \cong \overline{CB}$	Definition of congruence

Given the proof above, determine whether the following statements are true.

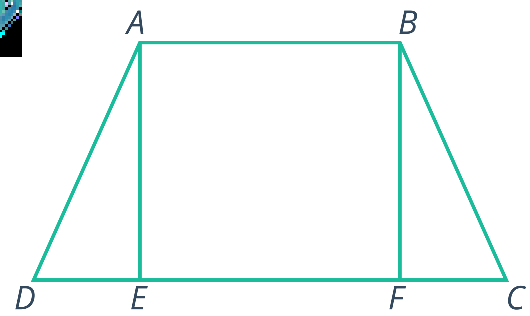
a If a trapezoid has congruent legs, then it is isosceles.

b If a trapezoid is isosceles, then it has congruent legs.

c If a trapezoid has one pair of congruent base angles, then it is isosceles.

d If a trapezoid is isosceles, then it has one pair of congruent base angles.

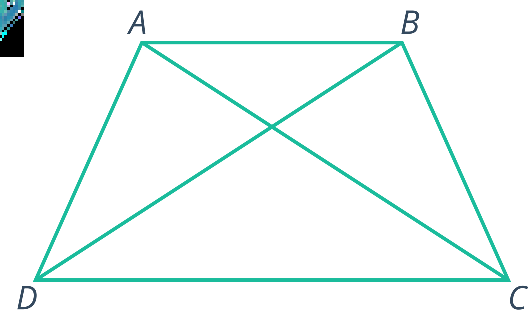
11 Given that $ABCD$ is an isosceles trapezoid with altitudes $\overline{AE}$ and $\overline{BF}$.
Complete the following two-column proof that proves that each pair of base angles are congruent.



To prove: Each pair of base angles of trapezoid $ABCD$ are congruent.

	Statements	Reasons
1.	$ABCD$ is an isosceles trapezoid	Given
2.	$\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3.	$\overline{AE} \cong \overline{BF}$	Altitudes of a trapezoid are congruent
4.	$\angle DEA$ and $\angle CFB$ are right angles	Definition of an altitude
5.	$\angle DEA \cong \angle CFB$	All right angles are congruent
6.		
7.		
8.	$\angle BCD$ and $\angle CBA$ are supplementary	Consecutive interior angles postulate
9.	$\angle ADC$ and $\angle BAD$ are supplementary	Consecutive interior angles postulate
10.	$\angle BAD \cong \angle ABC$	Congruent supplements theorem

12 Given that $ABCD$ is an isosceles trapezoid with bases $\overline{AB}$ and $\overline{DC}$.
Complete the following two-column proof that proves that $\overline{AC} \cong \overline{BD}$.



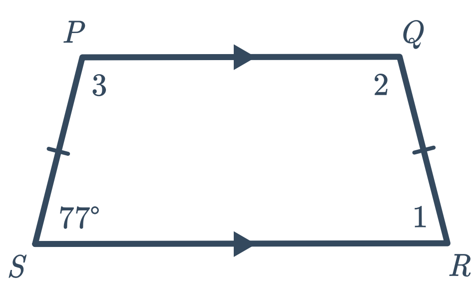
To prove: $\overline{AC} \cong \overline{BD}$

	Statements	Reasons
1.	$ABCD$ is an isosceles trapezoid	Given
2.	$\overline{AD} \cong \overline{BC}$	Definition of an isosceles trapezoid
3.		
4.	$\angle ADC \cong \angle BCD$	If a trapezoid is isosceles, then each pair of base angles is congruent.
5.	$\triangle ADC \cong \triangle BCD$	Side-angle-side congruence theorem
6.	$\overline{AC} \cong \overline{BD}$	Corresponding parts of congruent triangles are congruent (CPCTC)

Additional questions

13 Solve for each of the missing angle measures in the following trapezoids:

a



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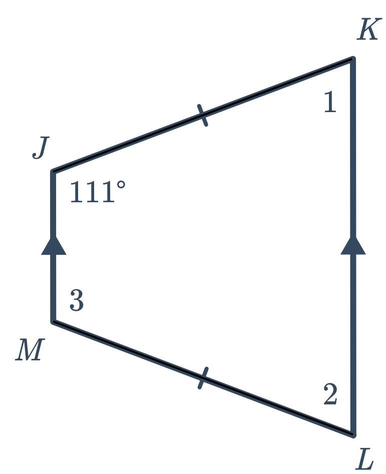

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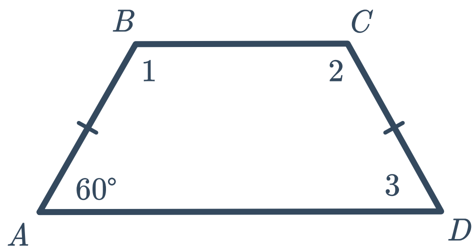
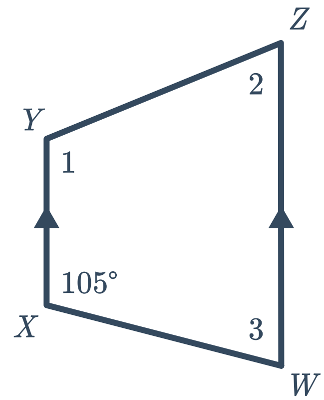
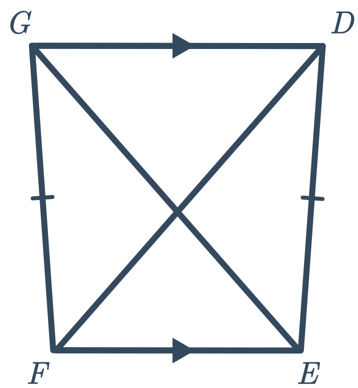


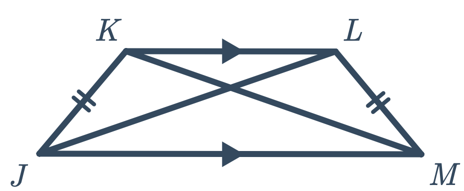


14 Given: Trapezoid $DEFG$
$EG = 3.5$



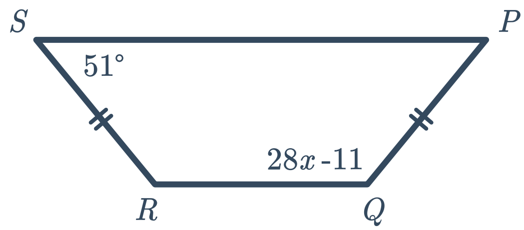
Solve for FD.

15 Given: Trapezoid $JKLM$
$KM = 1.26$



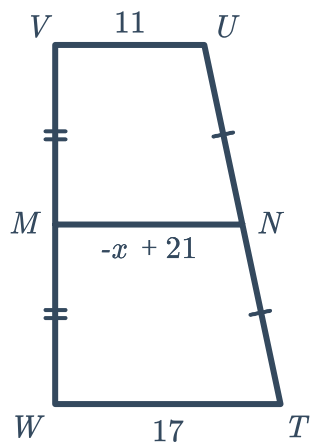
Solve for JL.

16 Given: Trapezoid $PQRS$



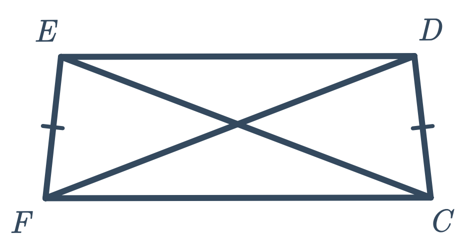
Solve for x.

17 Given: Trapezoid $TUVW$



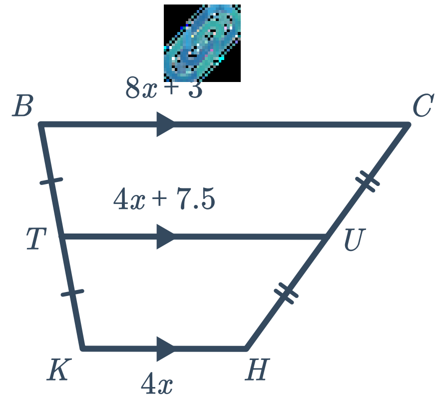
Solve for x.

18 Given: Trapezoid $FEDC$
$EC = 4x + 11$
$FD = 3x + 17$



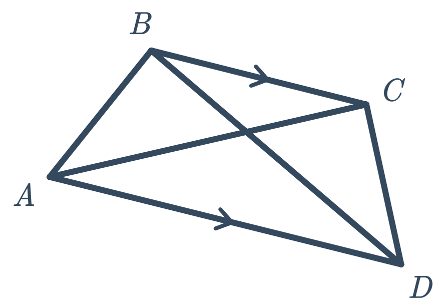
Solve for x.

19



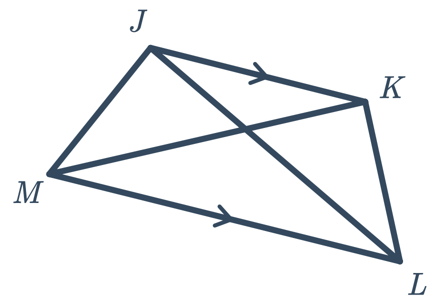
- **a** Solve for x
- **b** Solve for BC
- **c** Solve for TU
- **d** Solve for KH

20 Given: $\angle BAD = (5x^2+32)^\circ$
$\angle CDA = (7x^2)^\circ$



Find the value of x that makes Trapezoid $ABCD$ isosceles.

21 Determine whether each of the following st[redacted]ents about isosceles trapezoid $JKLM$ is true or false and explain how you know.



a $\overline{JM} \cong \overline{KL}$

b $\overline{JL} \cong \overline{KM}$

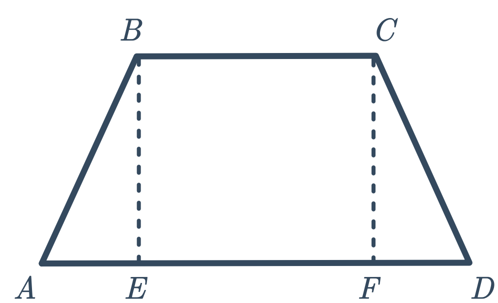
c $\overline{JK} \cong \overline{ML}$

d $\overline{JM} || \overline{KL}$

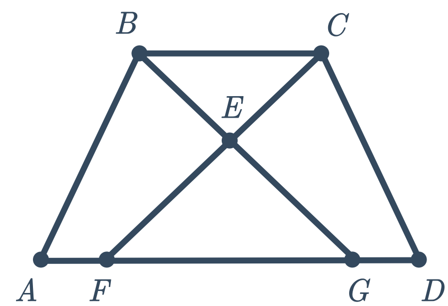
e $\angle J \cong \angle L$

f $\angle K$ and $\angle M$ are supplementary.

22 Given: Trapezoid $ABCD$ with altitudes $\overline{BE}$ and $\overline{CF}$
$\overline{AB} \cong \overline{CD}$
To prove: $\angle A \cong \angle D$



23 Given: Trapezoid $ABCD$
$\overline{EF} \cong \overline{EG}$
$\overline{AF} \cong \overline{DG}$
$\overline{BG} \cong \overline{CF}$
To prove: Trapezoid $ABCD$ is isosceles



24 Given: Trapezoid $ABCD$ is isosceles
$\angle BAK \cong \angle BKA$
To prove: $BKDC$ is a parallelogram